

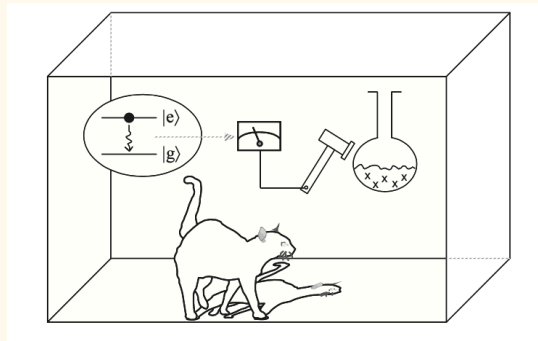
Quantum Information Applications

Schrodinger's Cat

Introduction: The Cat as an example

When I hear about Schrödinger's cat, I reach for my gun. – Stephen Hawking

Schrodinger's infamous cat faces life or death contingent upon an automatic device which breaks a vial of poison and kills the cat if an excited atomic state is observed to decay, as illustrated here:



Schrodinger's experiment asks "what happens if the atom is in a superposition state?". Is the cat alive or dead? It would seem that the cat's state is **entangled** with the state of the atom: if the atom is in the ground state, the cat will die. If the atom remains excited and does not decay, the cat is alive.

This would seem to indicate the cat is simultaneously alive and dead if the atom is in a superposition state. This scenario is entirely allowed by quantum mechanics...so why would this never actually happen in real life?

Quantum Information

In practice it is impossible to perfectly isolate the cat and the atom in their box, and thus information about this superposition state will leak into the external world. For example, heat from the cat's body could permeate the wall and give some indication of its state to the outside.

Manipulating Quantum Information

We know qubits can be continuously-valued, but despite this they are the smallest possible unit of quantum information. Confusingly, despite being the smallest unit of information, it is impossible to measure their value precisely. Five famous theorems describe the limits on manipulation of quantum information:

- *No-teleportation theorem*, which states that a qubit cannot be (wholly) converted into classical bits; that is, it cannot be fully "read".
- *No-cloning theorem*, which prevents an arbitrary qubit from being copied.
- *No-deleting theorem*, which prevents an arbitrary qubit from being deleted.
- *No-broadcast theorem*, which prevents an arbitrary qubit from being delivered to multiple recipients, although it can be transported from place to place (e.g. via quantum teleportation).
- *No-hiding theorem*, which demonstrates the conservation of quantum information.

2. Pick one of the theorems

(https://en.wikipedia.org/wiki/Quantum_information#Qubits_and_information_theory) And describe a consequence it has.

The Bekenstein bound

In the 1970s, physicist Jacob Bekenstein proposed an idea: there is a fundamental limit (bound) to how much information (or entropy) can be contained within a given finite region of space that has a finite amount of energy. Traditionally, we think of information as something abstract, but Bekenstein showed that physical laws constrain how densely information can be packed into space. His insight grew out of trying to preserve the second law of thermodynamics: that entropy never decreases.

Imagine you have a box. You might think you can store as much data inside as you want, limited only by your technology. Bekenstein argued otherwise: nature itself places a hard cap on how much information you can store in that box, depending on its energy and size.

Energy (E): How much mass-energy is inside the box.

Radius (R): How big the box is.

$2\pi k/hc$ are all constants!

$$S \leq \frac{2\pi k R E}{\hbar c}$$

The more energy and size you have, the more information you can store — but only up to a limit. If you tried to cram in too much, you would form a black hole, and the information would be hidden behind an event horizon. The bound gives a precise maximum to entropy S (which, in physics, is a measure of information or disorder)

In physics, entropy measures the number of microscopic ways a system can be arranged without changing what we see macroscopically. In information theory, more entropy means more possible bits (more ways to encode information). Thus, entropy and information are two sides of the same coin.

Questions

3. Explain in your own words why putting too much energy into a box can cause it to form a black hole.
4. Using the Bekenstein bound, estimate the maximum entropy you could store in a sphere of 1 meter radius containing 1 kilogram of energy. (You can either look up the constants if or just like true physicists set them all equal to 1!)
5. **(Challenge)** How does the Bekenstein bound suggest that surface area, not volume, is more fundamental when it comes to information? Does this suggest we can encode all the universe's information in 2D (holographic principle)?

Open Questions

In this class we learned about several paradoxes, problems, and phenomena that remain unexplained. Pick one of these problems/thought experiments that most interests you and write 2-3 sentences about it. (Just a small overview)

1. **The Galactic Center GeV Excess (GCE)** is an unexpected surplus of gamma-ray radiation in the center of the Milky Way Galaxy

2. **Boltzmann Brain:** Over a sufficiently long time, random fluctuations could cause particles to spontaneously form literally any structure of any degree of complexity, including a functioning human brain. (This is somewhat similar to the infinite monkey typing shakespeare thought experiment)
3. **Final parsec problem:** Supermassive black holes appear to have merged, and what appears to be a pair in this intermediate range has been observed, in PKS 1302–102. However, theory predicts that when supermassive black holes reach a separation of about one parsec, it may take billions of years to orbit closely enough to merge—greater than the age of the universe
4. **Quantum Entanglement and Nonlocality:** We know entangled particles influence each other instantly at a distance, seemingly faster than light. Entanglement is crucial for quantum communication, but no one fully agrees on what entanglement is ontologically.